

- For a relatively high gain and two complex impedances Z_1 and Z_2 :

$$\frac{V_{out}}{V_{in}} \approx -\frac{Z_1}{Z_2}$$

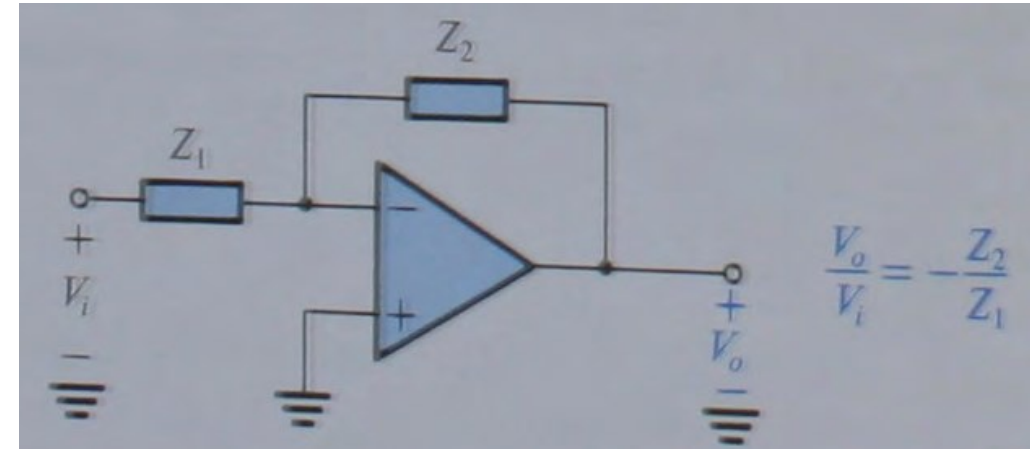
- If Z_1 is a resistor and Z_2 a capacitor:

$$Z_1 = (C_1 s)^{-1} \text{ and } Z_2 = R_1$$

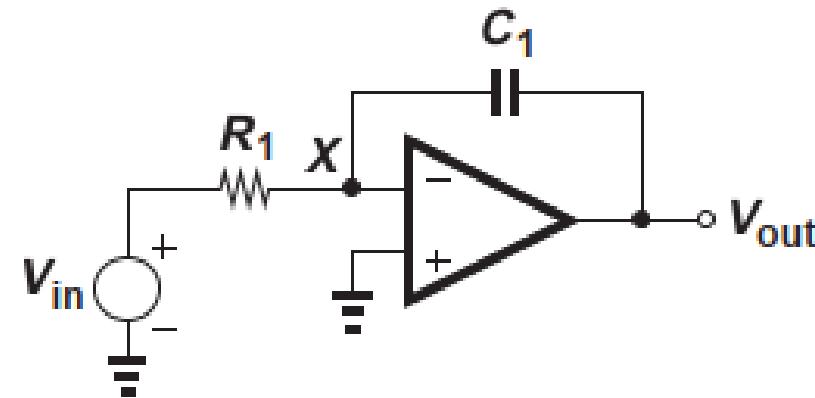
$$\frac{V_{out}}{V_{in}} = -\frac{1}{C_1 s R_1}$$
$$= -\frac{1}{R_1 C_1 s}$$

$$\frac{V_{in}}{R_1} = -C_1 \frac{dV_{out}}{dt}$$

$$V_{out} = -\frac{1}{R_1 C_1} \int V_{in} dt$$



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.



Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

Operational Amplifiers: (Inverting) Integrator

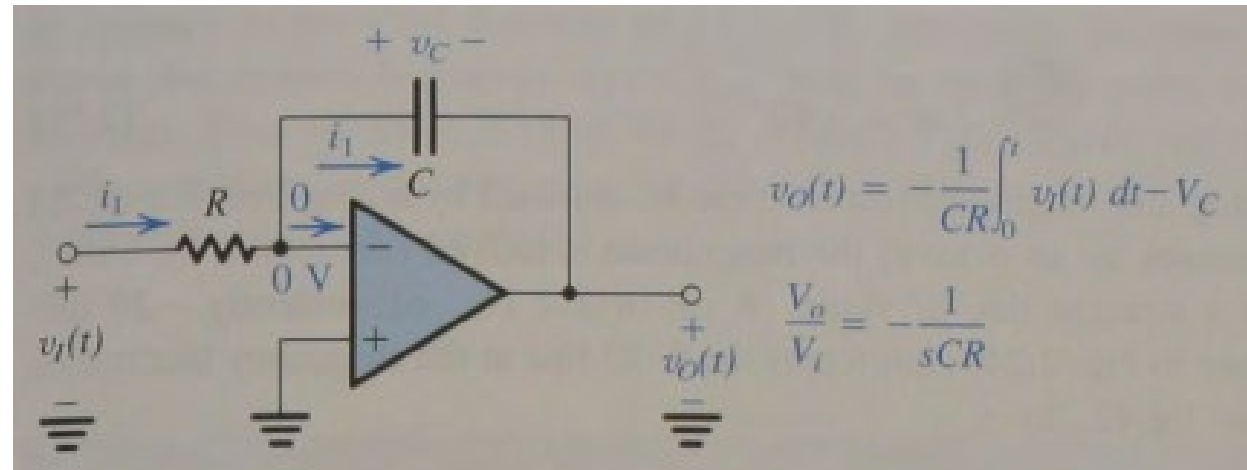
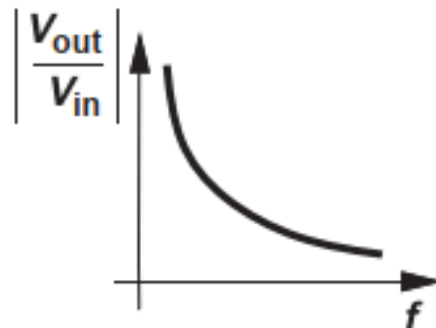
$$v_C(t) = V_C + \frac{1}{C} \int_0^t i_1(t) dt$$

$$v_O(t) = -\frac{1}{CR} \int_0^t v_I(t) dt - V_C$$

$$\frac{V_o(s)}{V_i(s)} = -\frac{1}{sCR}$$

$$\frac{V_o(j\omega)}{V_i(j\omega)} = -\frac{1}{j\omega CR}$$

$$\left| \frac{V_o}{V_i} \right| = \frac{1}{\omega CR}$$



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

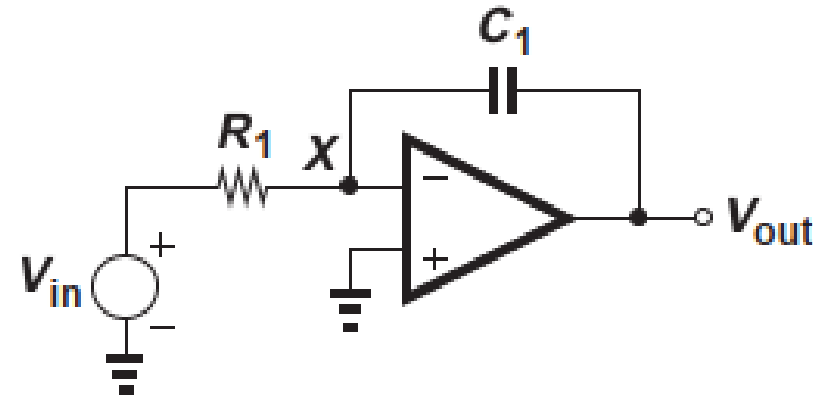
- For a regular gain:

$$\frac{V_{in} - V_X}{R_1} = \frac{V_X - V_{out}}{\frac{1}{C_1 s}}$$

$$V_X = \frac{V_{out}}{-A_0}$$

$$\frac{V_{out}}{V_{in}} = \frac{-1}{\frac{1}{A_0} + \left(1 + \frac{1}{A_0}\right) R_1 C_1 s}$$

$$s_p = \frac{-1}{(A_0 + 1) R_1 C_1}$$



Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

- For a relatively high gain and two complex impedances Z_1 and Z_2 :

$$\frac{V_{out}}{V_{in}} \approx -\frac{Z_1}{Z_2}$$

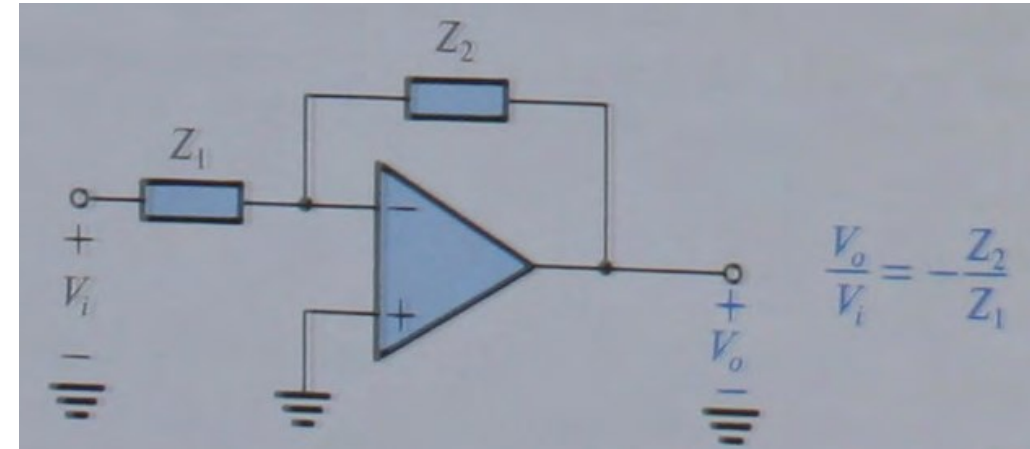
- If Z_1 is a capacitor and Z_2 a resistor:

$$\frac{V_{out}}{V_{in}} = -\frac{R_1}{\frac{1}{C_1 s}}$$

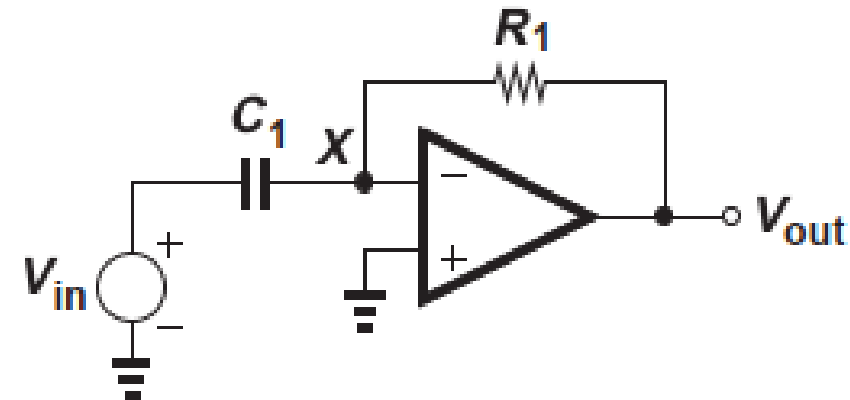
$$= -R_1 C_1 s$$

$$C_1 \frac{dV_{in}}{dt} = -\frac{V_{out}}{R_1}$$

$$V_{out} = -R_1 C_1 \frac{dV_{in}}{dt}$$



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.



Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

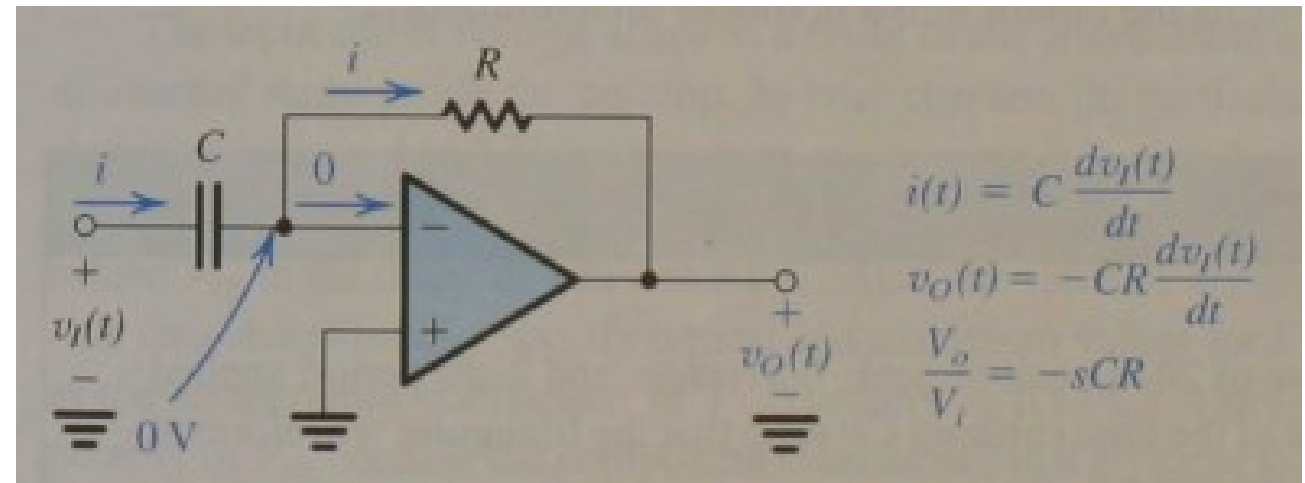
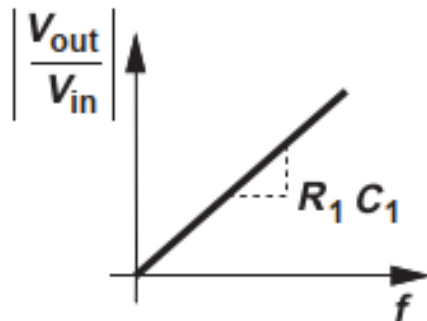
Operational Amplifiers: (Inverting) Differentiator

$$v_o(t) = -CR \frac{dv_i(t)}{dt}$$

$$\frac{V_o(s)}{V_i(s)} = -sCR$$

$$\frac{V_o(j\omega)}{V_i(j\omega)} = -j\omega CR$$

$$\left| \frac{V_o}{V_i} \right| = \omega CR$$



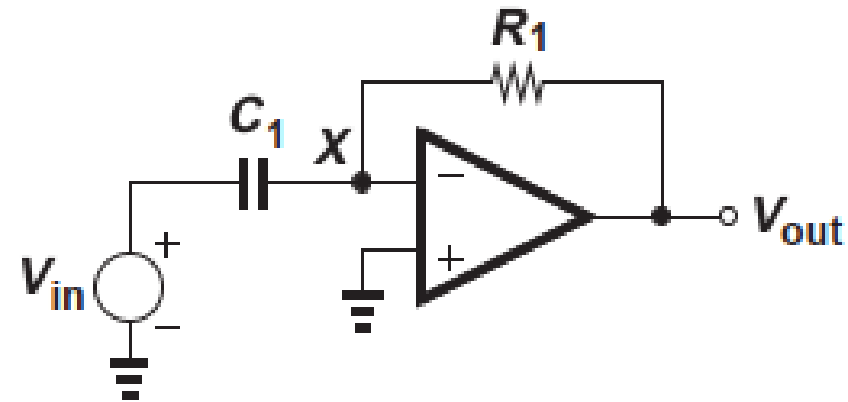
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

- For a regular gain:

$$\frac{V_{in} - V_X}{\frac{1}{C_1 s}} = \frac{V_X - V_{out}}{R_1}$$

$$\frac{V_{out}}{V_{in}} = \frac{-R_1 C_1 s}{1 + \frac{1}{A_0} + \frac{R_1 C_1 s}{A_0}}$$

$$s_p = -\frac{A_0 + 1}{R_1 C_1}$$

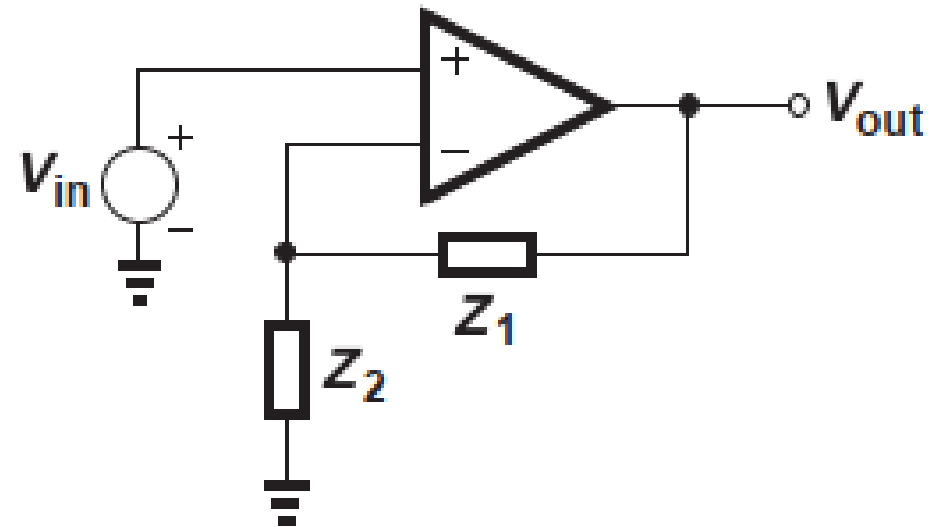


Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

- For a relatively high gain and two complex impedances Z_1 and Z_2 :

$$\frac{V_{out}}{V_{in}} = 1 + \frac{Z_1}{Z_2}$$

- Non-inverting cases



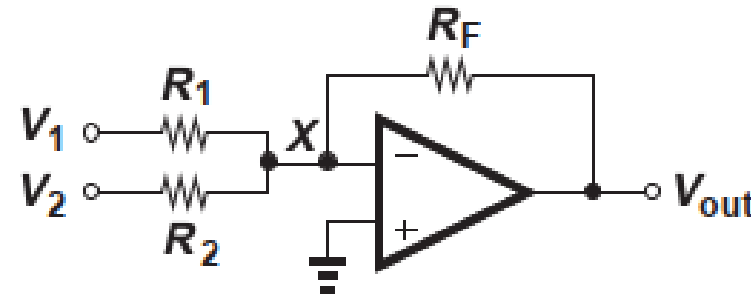
Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} = \frac{-V_{out}}{R_F}$$

$$V_{out} = -R_F \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

- For $R_1 = R_2 = R$:

$$V_{out} = \frac{-R_F}{R} (V_1 + V_2)$$



Razavi: "Fundamentals of Microelectronics", 3rd edition, Wiley & Sons, 2021.

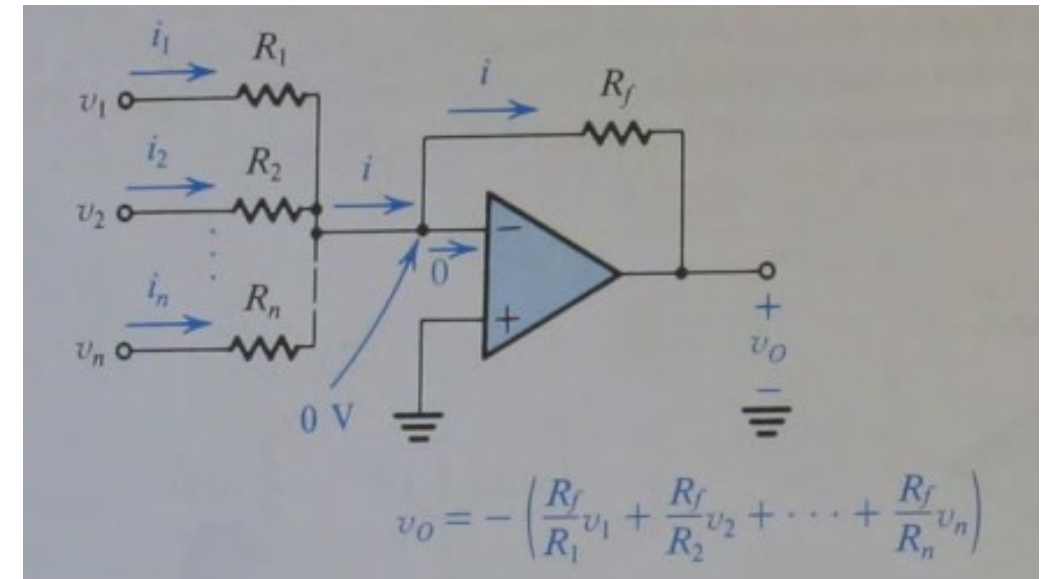
Operational Amplifiers: Voltage Adder / Weighted Summer

$$i_1 = \frac{v_1}{R_1}, \quad i_2 = \frac{v_2}{R_2}, \quad \dots, \quad i_n = \frac{v_n}{R_n}$$

$$i = i_1 + i_2 + \dots + i_n$$

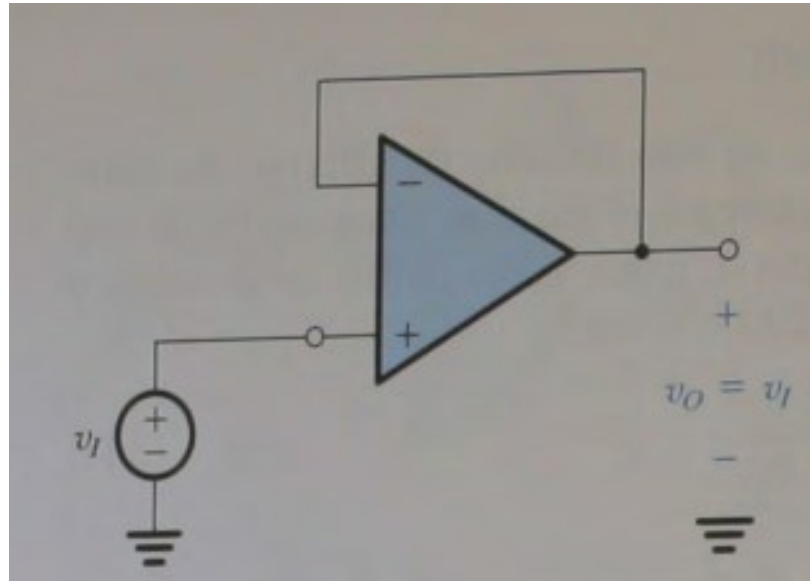
$$v_O = 0 - iR_f = -iR_f$$

$$v_O = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \dots + \frac{R_f}{R_n}v_n\right)$$



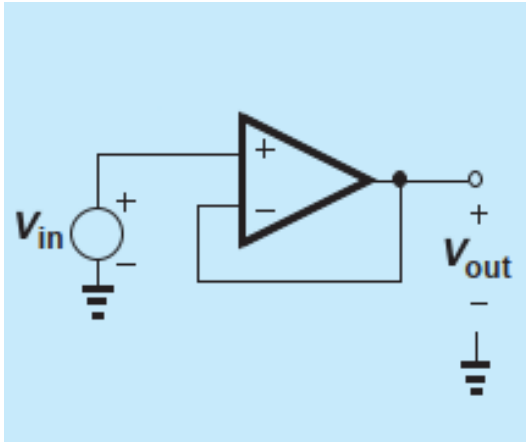
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Operational Amplifiers: Voltage Follower / Buffer



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

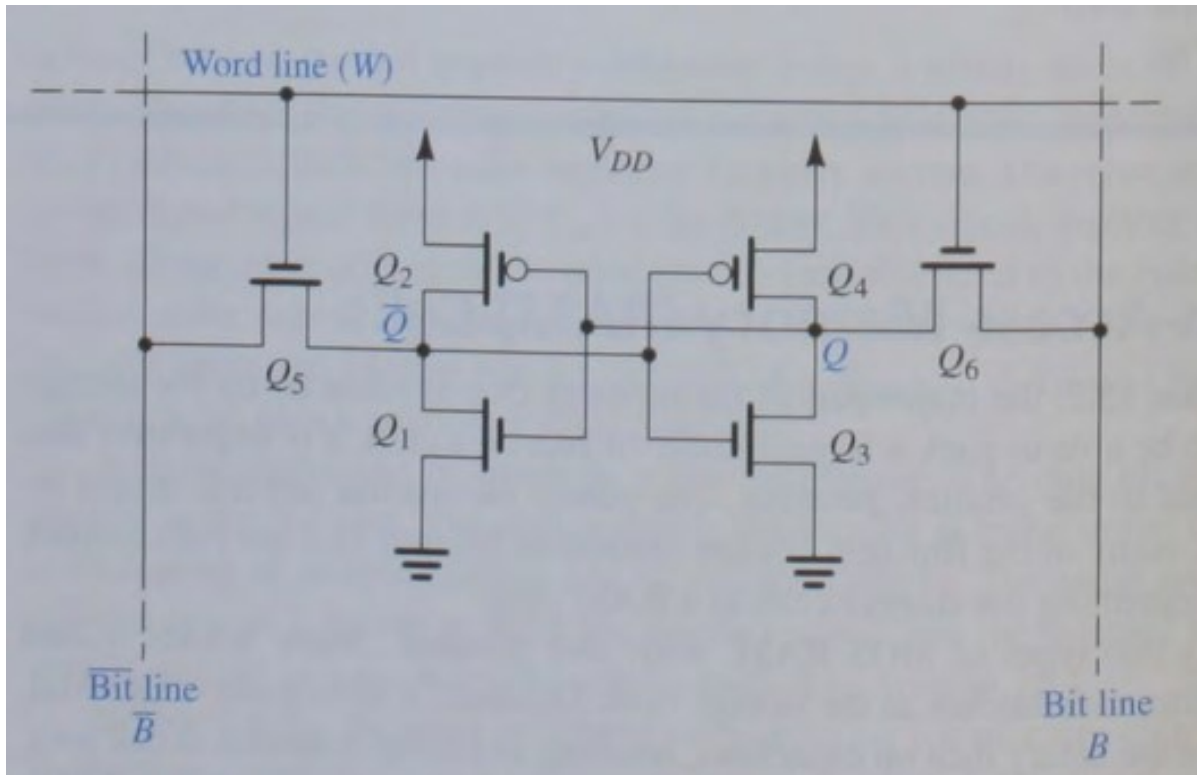
Operational Amplifiers: Voltage Follower / Buffer



$$V_{out} = A_0(V_{in1} - V_{in2}) = A_0(V_{in} - V_{out})$$

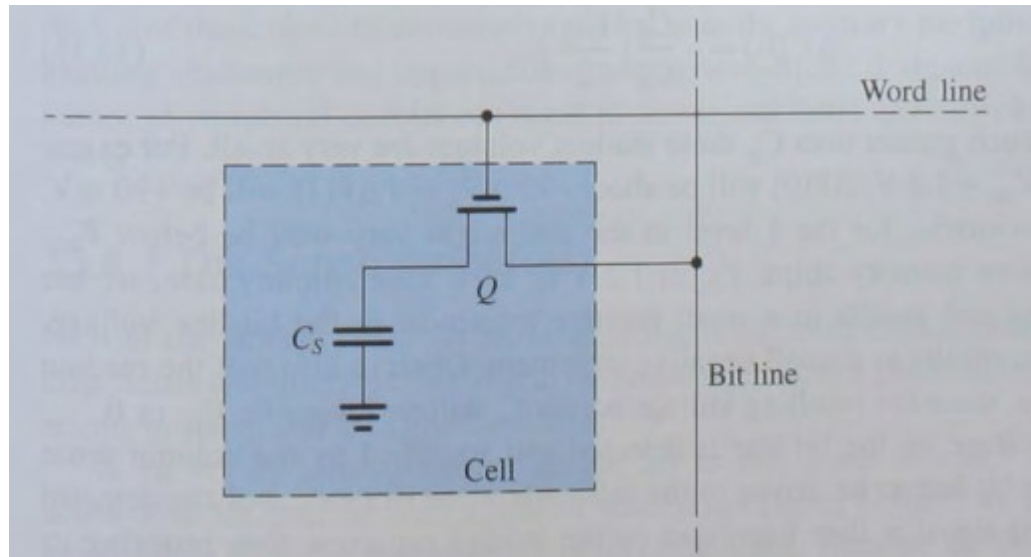
$$\frac{V_{out}}{V_{in}} = \frac{A_0}{1 + A_0}$$

Razavi: “Fundamentals of Microelectronics”, 3rd edition, Wiley & Sons, 2021.



- Static Random-Access Memory (SRAM) Cell

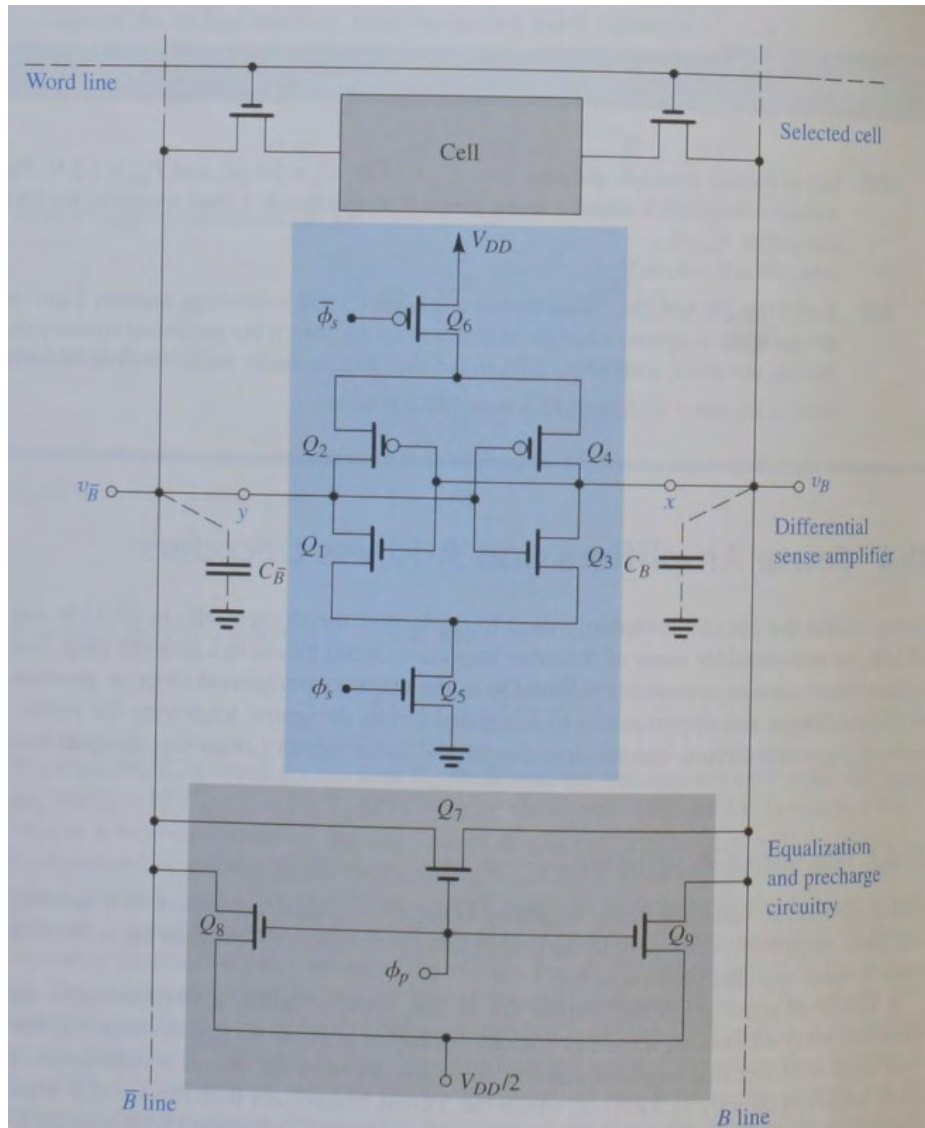
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.



- Dynamic Random-Access Memory (DRAM) Cell

Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

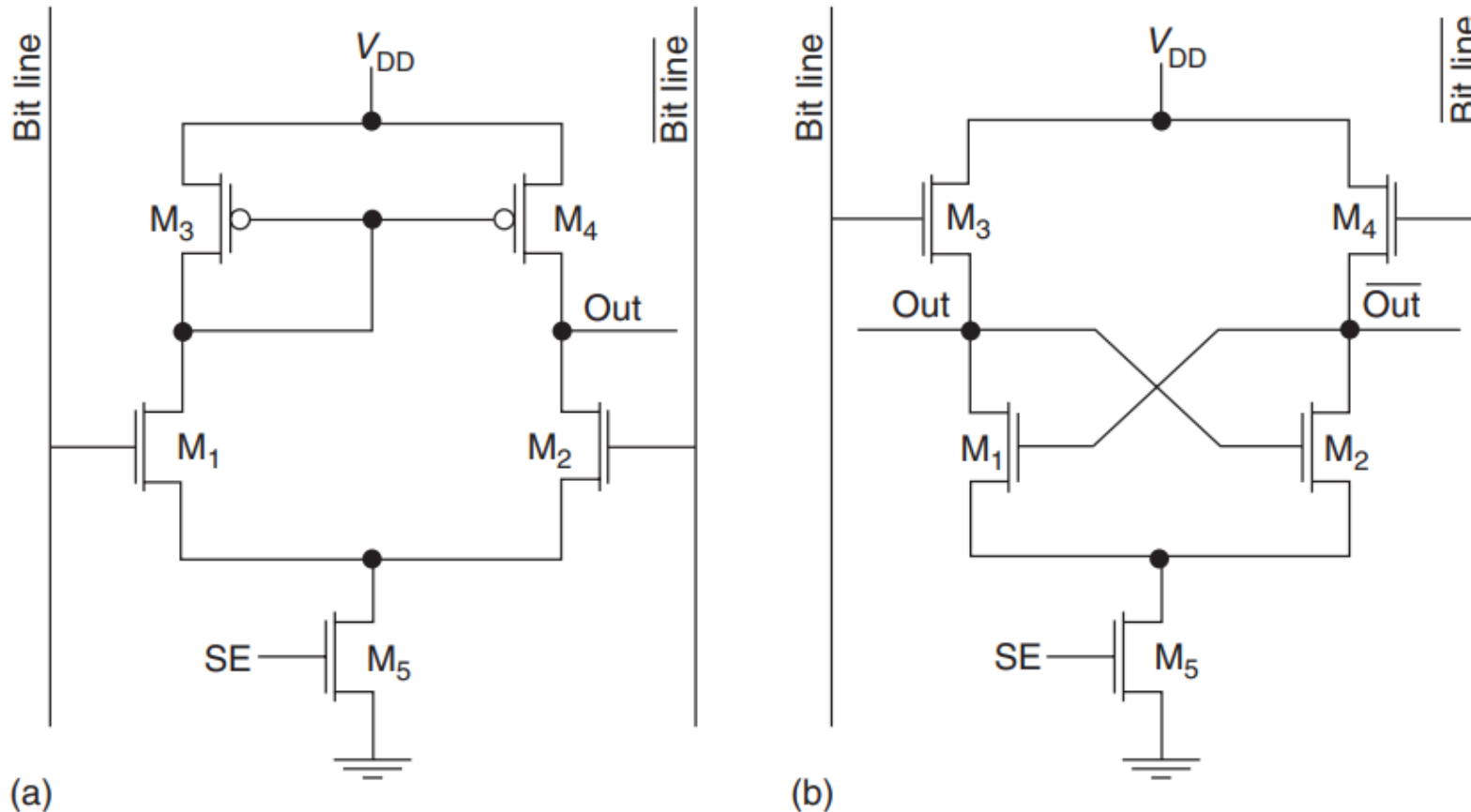
Basic Digital Circuits: Memories (Sense Amplifiers)



- Sense amplifiers (for memory content acquisition)
- Equalisation and precharging

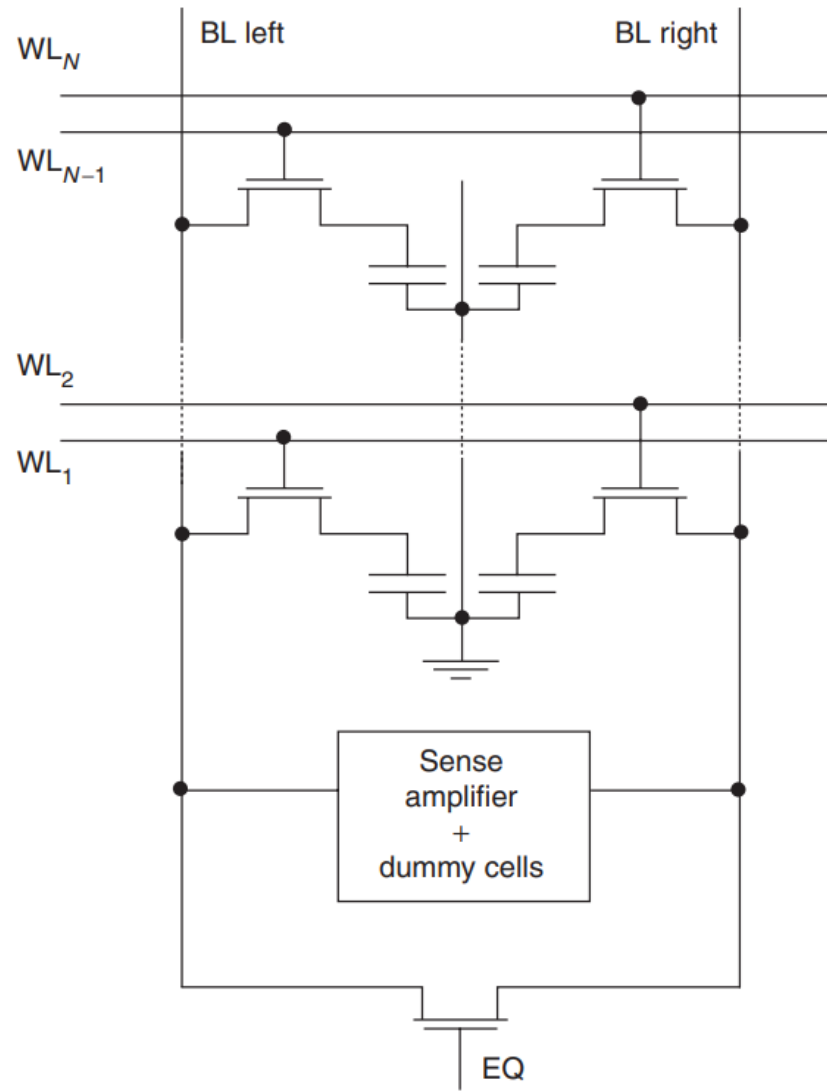
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Basic Digital Circuits: Memories (Sense Amplifiers)

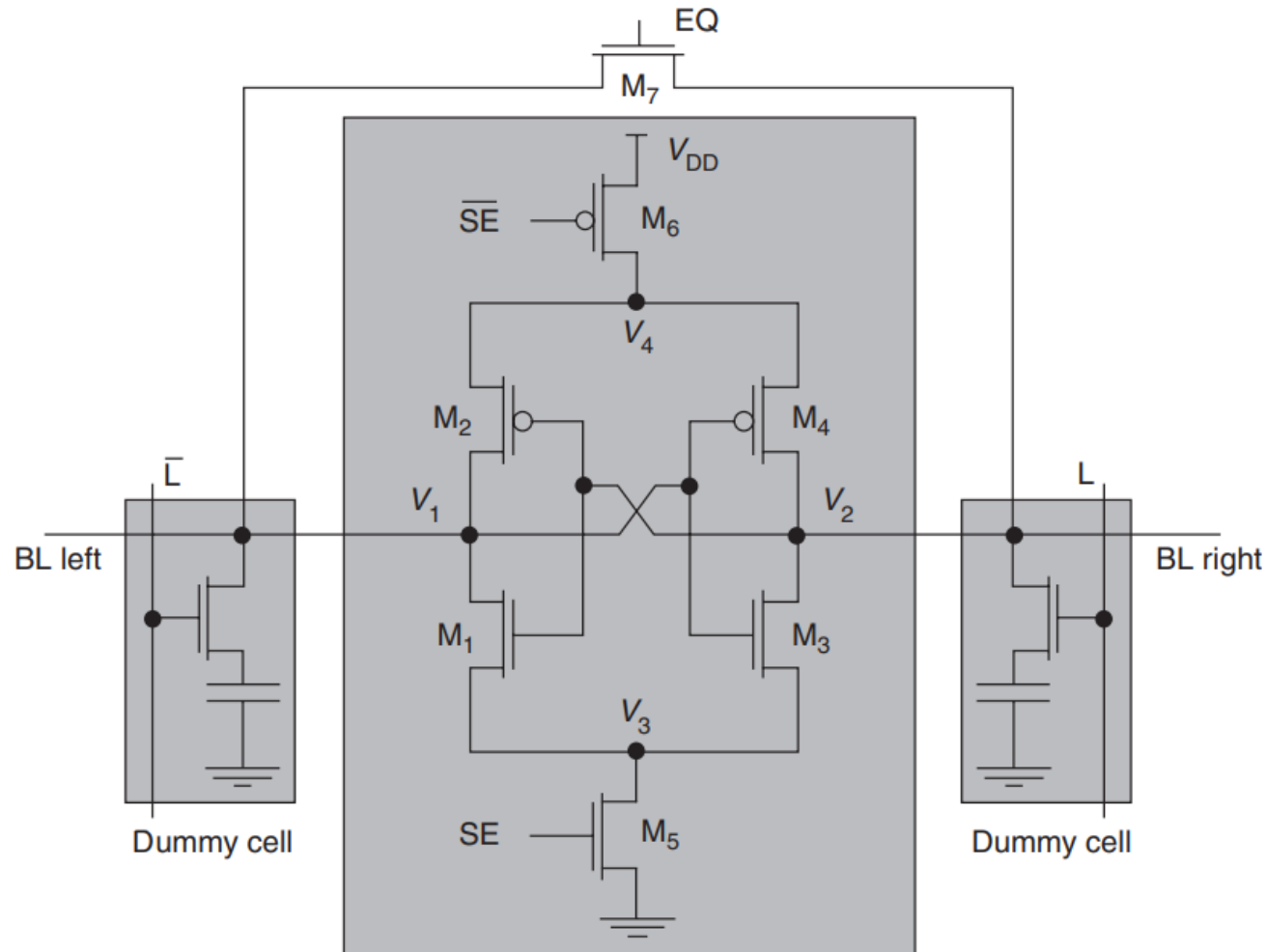


SRAM sense amplifier. (a) voltage amplifier, (b) level shifter.

- SRAM sense amplifier



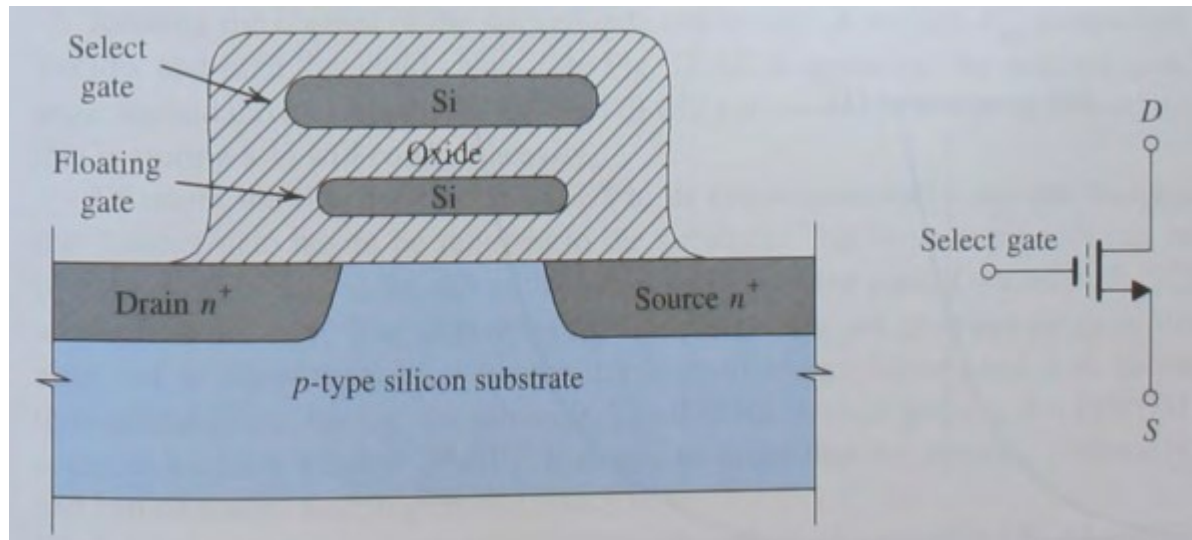
- DRAM organisation



- DRAM sense amplifier

Schematic depiction of the DRAM sense amplifier.

Baccarani, G., & Gnani, E. (2005). Memory Devices, Volatile. Encyclopedia of Condensed Matter Physics, 337–347.



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

- Programmable Read-Only Memory (ROM)
- Electrically Erasable/Programmable ROM (Flash memory)